

Skin Cancer Analysis – Full Insights Report

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Executive Summary

This report presents a comprehensive analysis of demographic, clinical, environmental, and lifestyle factors associated with skin cancer diagnoses. Using advanced SQL techniques, the dataset was examined to uncover patterns in severity, recurrence, and risk.

Key findings include:

- Middle-aged adults and males show the highest diagnosis rates
- The face is the most affected region
- Actinic keratosis (ACK) is the most common diagnosis
- Melanoma (MEL) has the largest lesion diameters
- Smoking and drinking significantly increase the likelihood of severe diagnoses
- Poor sanitation does not strongly correlate with severity

These insights support targeted prevention, early detection, and lifestyle-focused interventions.

1. Introduction

1.1 Background

Skin cancer is one of the most common malignancies worldwide. Understanding the demographic, environmental, and behavioural factors that influence diagnosis and severity is essential for effective prevention and treatment.

1.2 Objectives

This analysis aims to:

- Identify high-risk demographic groups
- Analyse lesion characteristics and severity indicators
- Evaluate environmental and sanitation influences
- Assess lifestyle behaviours and their relationship to severe diagnoses

2. Demographic Insights

2.1 Age-Related Risk

Middle-aged adults (50–64) represent the largest share of diagnoses. This aligns with known patterns: cumulative UV exposure increases with age, and early signs of skin damage often manifest in mid-life.

Implication: This age group should be prioritised for screening campaigns and educational outreach.

2.2 Gender Distribution

Males show a higher number of diagnoses than females.

Possible explanations:

- Higher occupational UV exposure
- Lower sunscreen use
- Lower likelihood of early dermatology visits

Implication: Public health messaging should be tailored to men, emphasising early detection and protective behaviours.

2.3 History of Skin Cancer

A notable number of patients have a previous history of skin cancer.

Interpretation: Skin cancer recurrence is common. Patients with prior diagnoses should be monitored closely and encouraged to attend regular check-ups.

3. Lesion Characteristics & Diagnosis Patterns

3.1 Most Common Diagnosis

Actinic keratosis (ACK) is the most frequent diagnosis in the dataset. This indicates widespread chronic sun damage in the population.

3.2 Lesion Growth & Symptoms

The most common symptoms reported were:

- Growth
- Itching
- Elevation

These symptoms are early warning signs of malignant transformation.

Interpretation: Clinicians should treat these symptoms as red flags and prioritise biopsy or dermoscopic evaluation.

3.3 Biopsy Trends

The face has the highest number of biopsied lesions, likely due to:

- High UV exposure
- High visibility of changes
- Patient concern about facial lesions

3.4 Lesion Diameter

Melanoma (MEL) shows the largest average lesion diameter.

Interpretation: Melanomas tend to grow more aggressively. Large diameter should trigger urgent referral and intervention.

4. Environmental & Sanitation Insights

4.1 Diagnoses by Body Region

The face is the most affected region across all diagnosis types.

Interpretation: UV exposure remains the dominant environmental risk factor. This reinforces the need for sunscreen use, hats, shade, and behavioural education.

4.2 Sanitation Access

A significant portion of patients lack piped water or sewage systems. However, **poor sanitation does not correlate strongly with severe diagnoses.**

Interpretation: Skin cancer severity is not driven by sanitation conditions. UV exposure, genetics, and lifestyle behaviours are more influential.

5. Lifestyle & Behavioural Insights

5.1 Smoking and Drinking Prevalence

- 62 smokers
- 138 drinkers
- 28 patients both smoke and drink

This provides a strong base for behavioural risk analysis.

5.2 Diagnosis Patterns Among Smokers & Drinkers

Basal Cell Carcinoma (BCC) is the most common diagnosis among:

- Smokers
- Drinkers
- Patients who both smoke and drink

Interpretation: Lifestyle factors may accelerate UV-related skin damage and impair immune response.

5.3 Risk Ratios

- Smokers are **2.6× more likely** to develop severe diagnoses
- Drinkers are **2.72× more likely**
- Patients who both smoke and drink: **82%** have malignant conditions

Interpretation: Alcohol shows a slightly stronger association with severity than smoking. Combined behaviours dramatically increase risk.

6. Integrated Insights

Bringing all factors together:

- **UV exposure** is the strongest environmental driver
- **Lifestyle behaviours** significantly amplify severity
- **Demographics** (age, gender) shape vulnerability
- **Lesion characteristics** (growth, diameter) predict malignancy
- **Sanitation** has minimal impact

This multi-factor view supports targeted interventions.

7. Recommendations

7.1 Public Health

- Focus prevention campaigns on middle-aged adults and men
- Promote face-focused UV protection
- Encourage regular skin checks

7.2 Clinical Practice

- Treat itching, elevation, and growth as early warning signs
- Prioritise biopsy for facial lesions
- Monitor patients with prior skin cancer closely

7.3 Lifestyle Interventions

- Integrate smoking and alcohol reduction into dermatology care
- Provide risk-communication materials showing severity associations

7.4 Research & Data

- Collect data on sun exposure, occupation, sunscreen use
- Explore combined UV + lifestyle risk pathways

8. Limitations

- Dataset may not represent the general population
- Missing variables (family history, UV exposure) limit modelling
- Boolean symptom fields reduce granularity

9. Conclusion

This analysis reveals clear demographic, clinical, and behavioural patterns in skin cancer diagnoses. The findings highlight the importance of UV protection, early detection, and lifestyle modification. The insights can guide public health strategies, clinical decision-making, and future research.